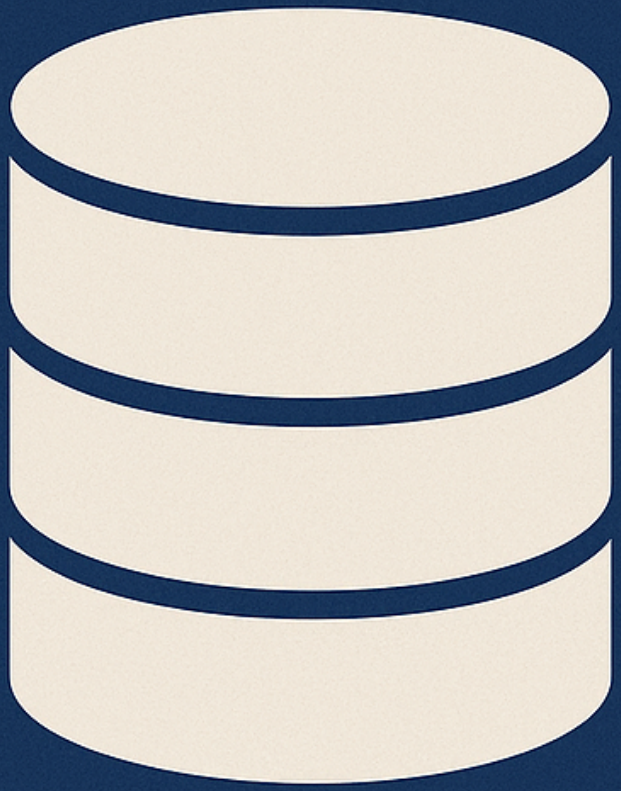
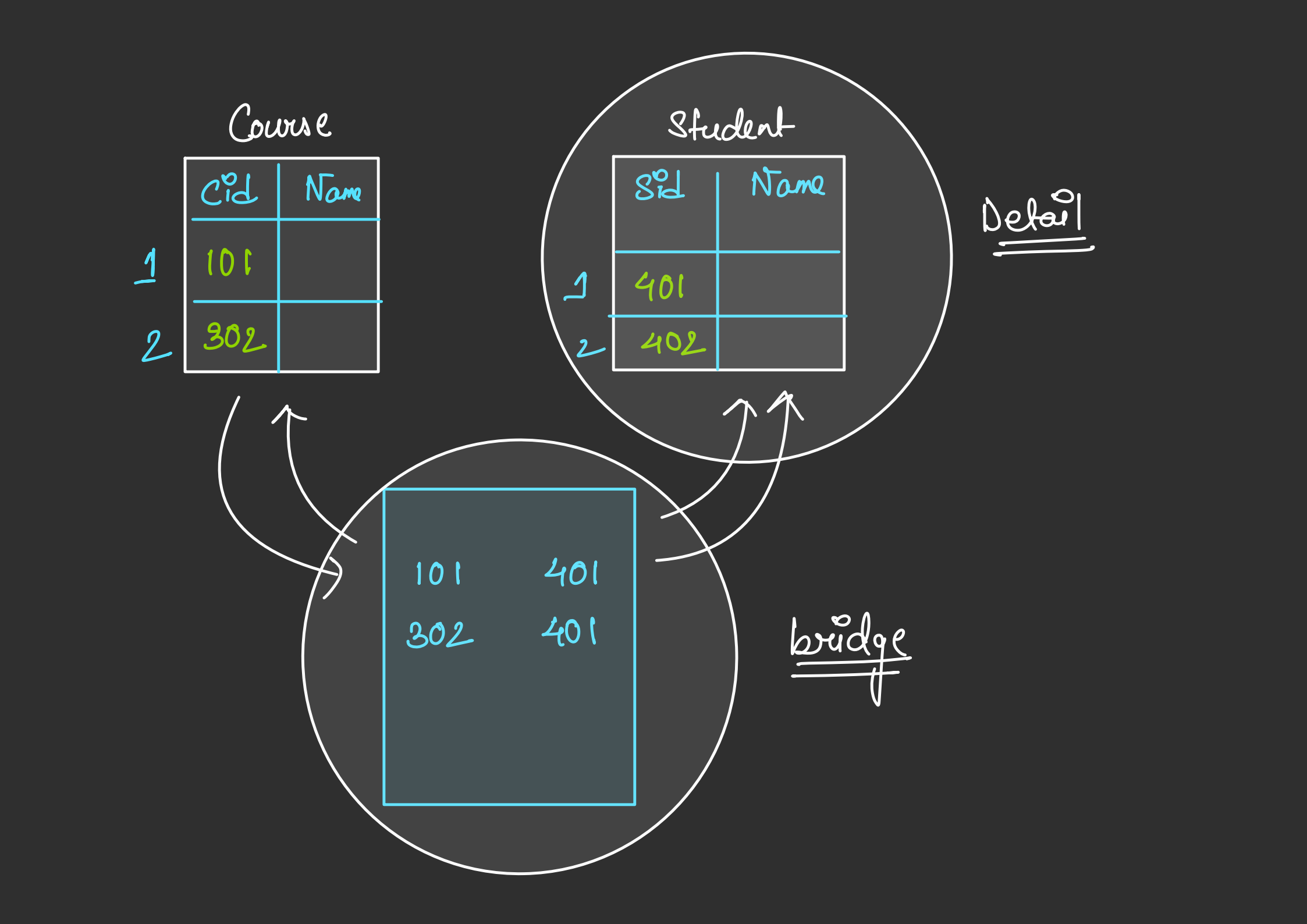
SQL

Class starts
@ 9:10pm



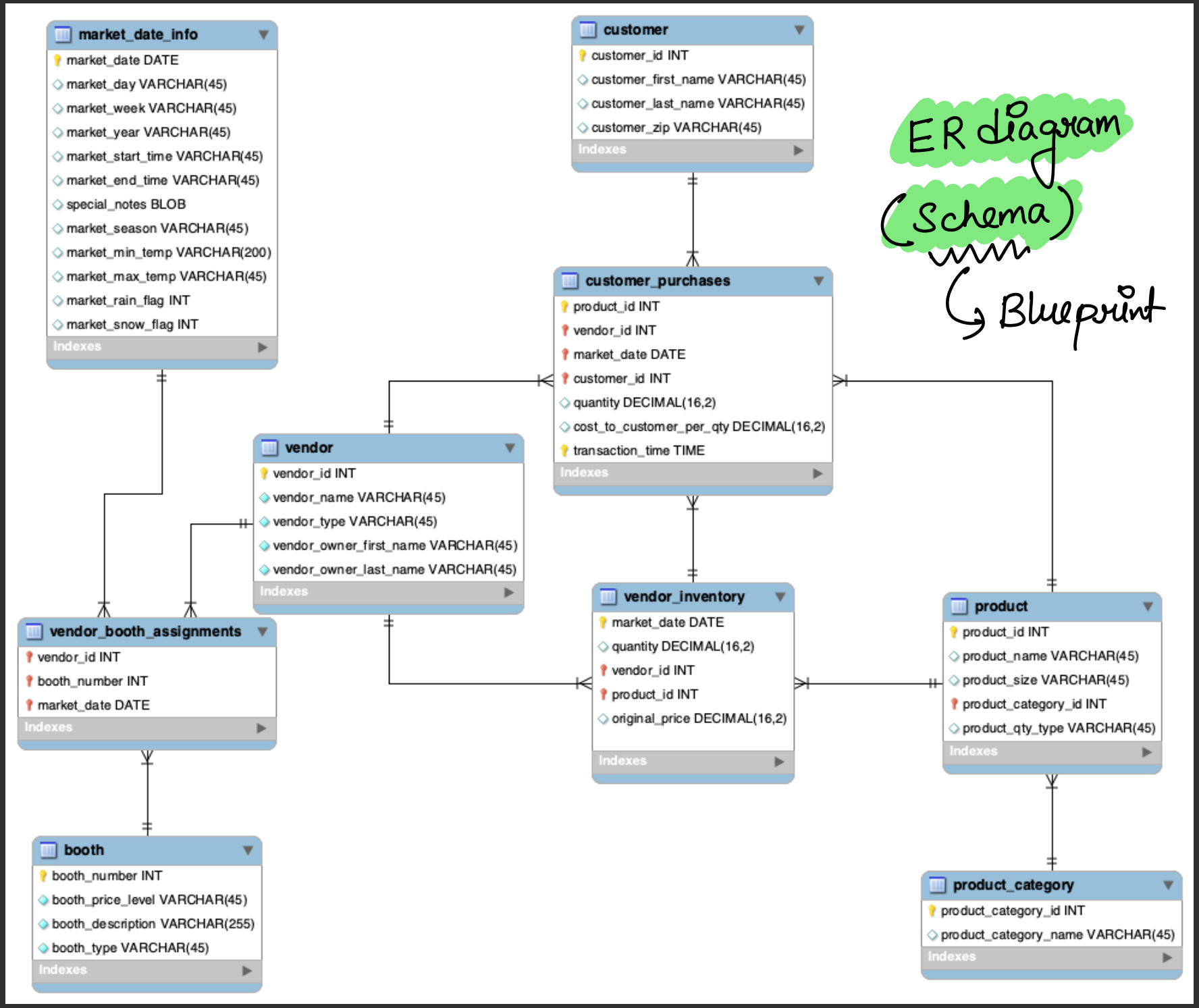



Course
Cid
Name
1
101
2
302
Student
Sid
Name
1
401
2
402
Detail
101
401
302
401
bridge

Lecture 4: Conditional Logic

#Agenda

1) DISTINCT → No duplicate
2) Subqueries
3) Case Statements
4) Data aggregating (grouping of data)


market_date_info
market_date DATE
market_day VARCHAR(45)
market_week VARCHAR(45)
market_year VARCHAR(45)
market_start_time VARCHAR(45)
market_end_time VARCHAR(45)
special_notes BLOB
market_season VARCHAR(45)
market_min_temp VARCHAR(200)
market_max_temp VARCHAR(45)
market_rain_flag INT
market_snow_flag INT
Indexes
customer
customer_id INT
customer_first_name VARCHAR(45)
customer_last_name VARCHAR(45)
customer_zip VARCHAR(45)
Indexes
ER diagram
(Schema)
Blueprint
customer_purchases
product_id INT
vendor_id INT
market_date DATE
customer_id INT
quantity DECIMAL(16,2)
cost_to_customer_per_qty DECIMAL(16,2)
transaction_time TIME
Indexes
vendor
vendor_id INT
vendor_name VARCHAR(45)
vendor_type VARCHAR(45)
vendor_owner_first_name VARCHAR(45)
vendor_owner_last_name VARCHAR(45)
Indexes
vendor_booth_assignments
vendor_id INT
booth_number INT
market_date DATE
Indexes
vendor_inventory
market_date DATE
quantity DECIMAL(16,2)
vendor_id INT
product_id INT
original_price DECIMAL(16,2)
Indexes
product
product_id INT
product_name VARCHAR(45)
product_size VARCHAR(45)
product_category_id INT
product_qty_type VARCHAR(45)
Indexes
booth
booth_number INT
booth_price_level VARCHAR(45)
booth_description VARCHAR(255)
booth_type VARCHAR(45)
Indexes
product_category
product_category_id INT
product_category_name VARCHAR(45)
Indexes

Distinct

ID	Name
1	Aryan
2	Amar
3	Akash
5	Aryan
2	Amar

distinct →

Result

ID	Name
1	Aryan
2	Amar
3	Akash
5	Aryan

Code

```
Select
distinct ID, Name
from
TableNam
```

Null Values

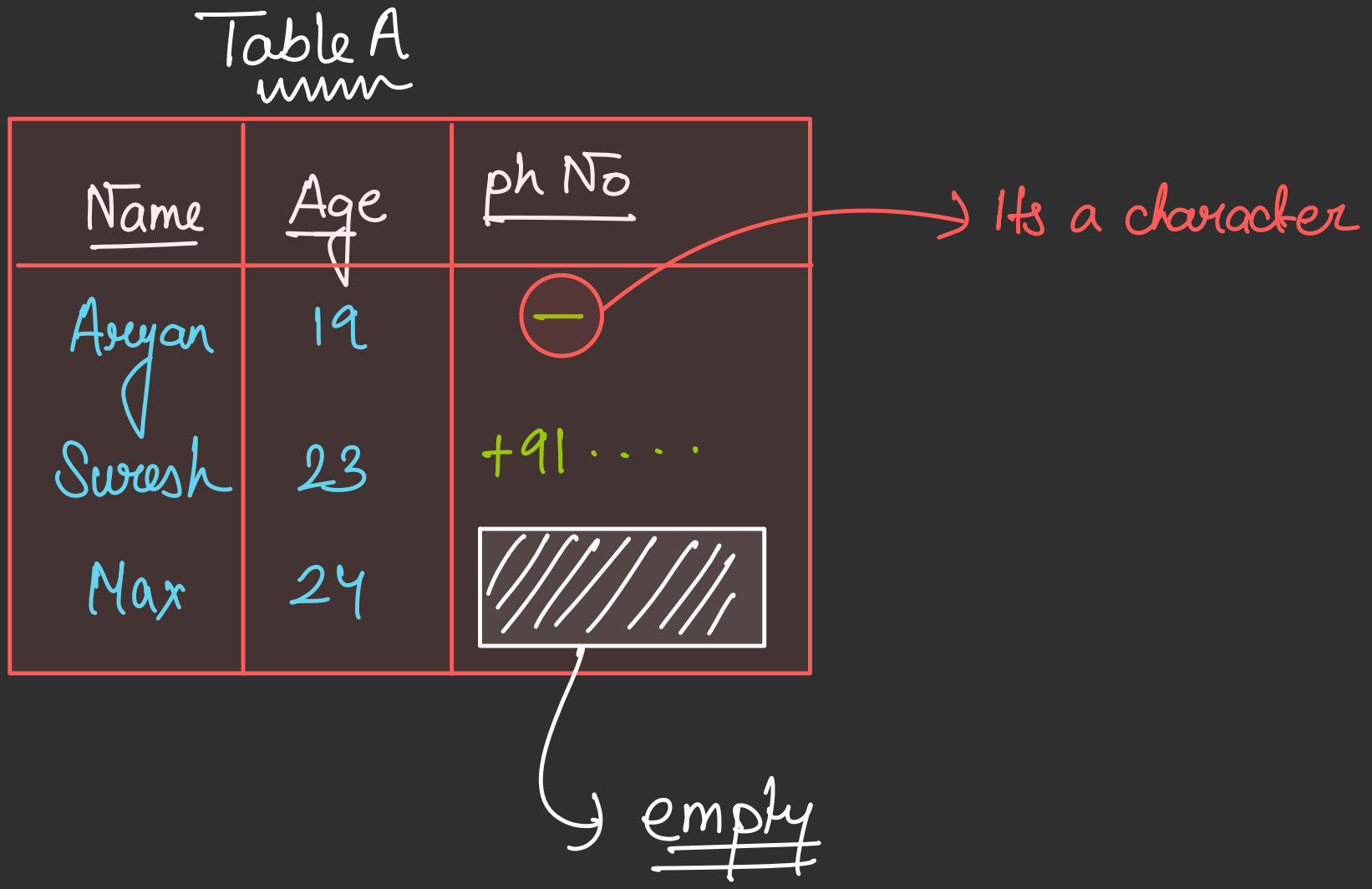
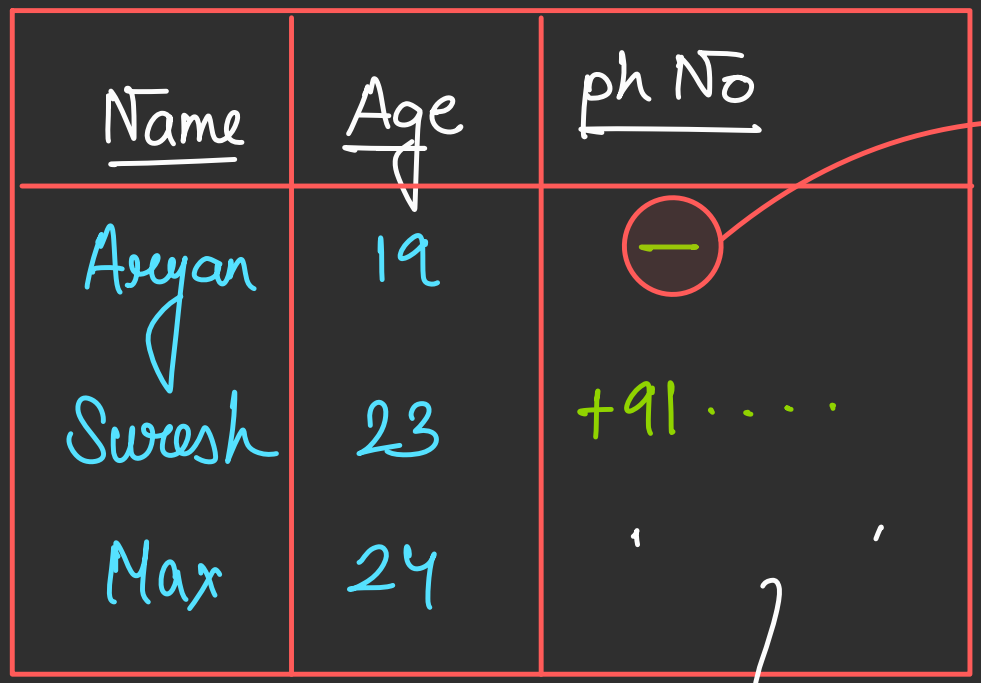

Table A
Name
Age
ph No
Aryan
19
Suresh
23
+91....
Max
24
Its a character
empty

Table A

Name	Age	ph No
Aryan	19	—
Suresh	23	+91....
Max	24	

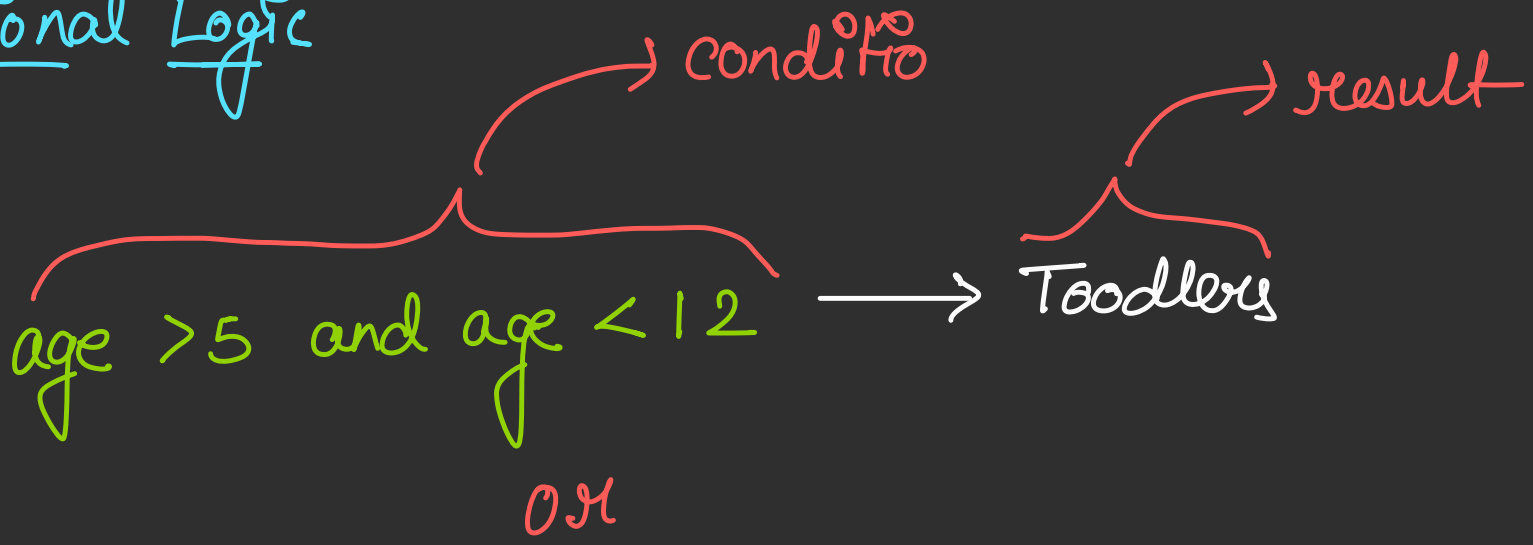


size is Null ⟶ will give you only null values when used with where

size = "null" ⟶ will give result on when the coloumn value has text as "null"

size = null ✗ (Error, Not a valid syntax)

Conditional Logic



Case Statement

```
Select a, b, c, d . . . ,
  CASE
    WHEN age >5 and age <12 THEN "Toodlers"
    WHEN age >13 and age <19 THEN "Teen"
    ELSE "Adults"
  END
From
  Table
```

#If function

Syntax

IF (condition , true value , false value)

Example : IF (product_size = "small" ,
"Yes, it is small" ,
"No, it's not small")

IF (condition,

true value,

IF (condition,

true

false)

false value)

Data Aggregation

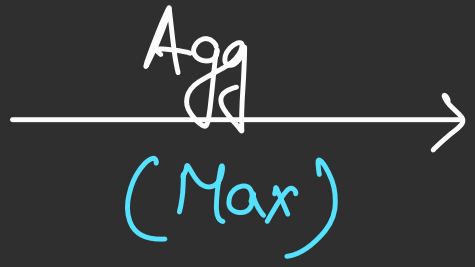
↳ putting things together / grouping.

"Compressing data" is how compression

Table

Col A
13
16
14
12
19

Rows = 5



Col A
19

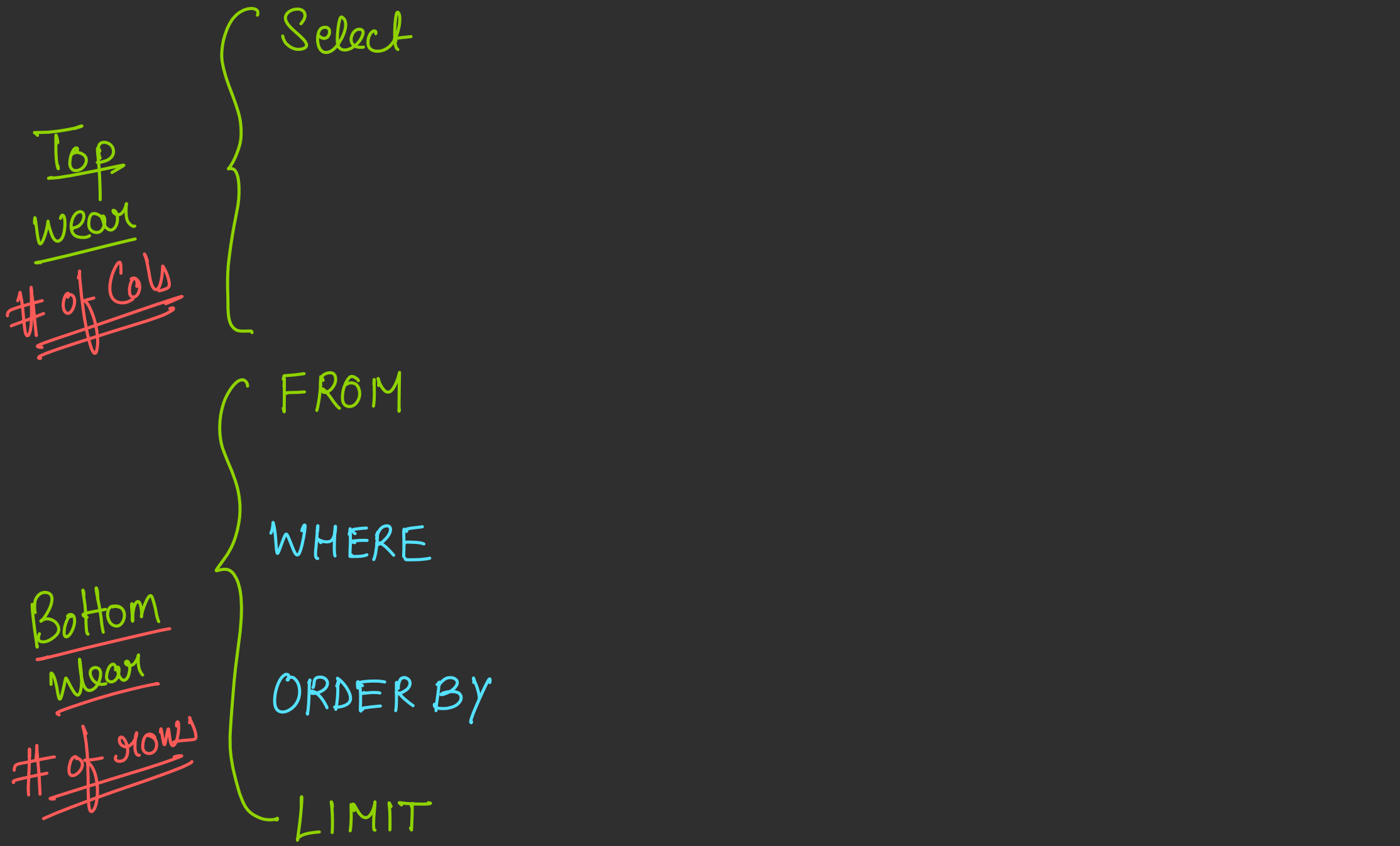
Row = 1

filtering row

(where, ...)

Aggregation

(MAX, MIN, AVG, distinct count)


Select
Top wear
of Cols
FROM
WHERE
ORDER BY
LIMIT
Bottom wear
of rows